

CELLAIR Mesoscope

CELL Analysis with Intelligent Recognition

Florian Feigl

A rectangular box containing the text "University Logo".

University
Logo

University of Salzburg

March 31, 2026

What We Observe

Calcium production through cell–cell interaction

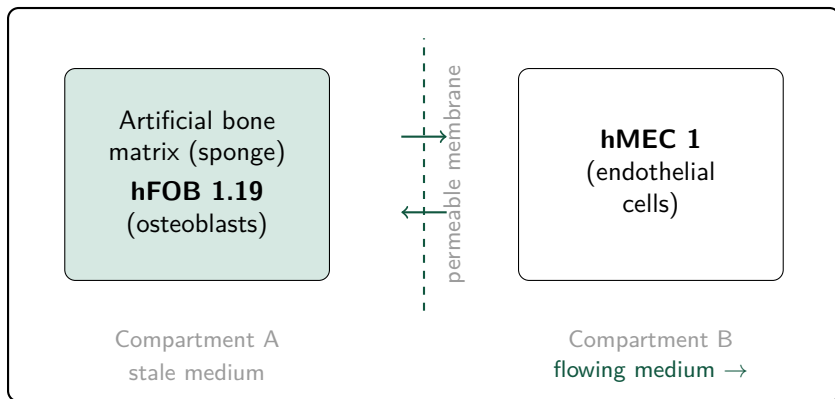
- ▶ **hFOB 1.19** (human fetal osteoblasts) and **hMEC 1** (human microvascular endothelial cells)
- ▶ Osteoblasts produce **hydroxyapatite** ($\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$) — the mineral component of bone
- ▶ Key question: How do endothelial cells influence and respond to osteoblast mineralization?

Relevant for bone tissue engineering, implant osseointegration, and regenerative medicine.



Technical Approach — Simulating the Natural Environment

Bioreactor Chip — 37 °C incubator



Flow rate: TBD (depends on chip geometry and shear stress tolerance)



Current Status — What We Have

General experiment structure established.

Bringing specimen and observation unit in place for early stage evaluation.

Component	Status
Bioreactor chip design	Established
Cell lines (hFOB 1.19, hMEC 1)	Available
Computing (RPi 5 + Hailo-8 NPU)	Operational
HQ Camera (IMX477, C-mount)	Acquired
OpenFlexure server (v3)	Running

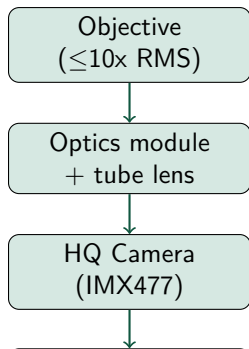
Currently verifying imaging quality on conventional microscope before transitioning to OpenFlexure platform.



OpenFlexure Microscope — Hardware Platform

Open-source, 3D-printed motorised microscope

Cambridge, CERN-OHL license



Pending hardware:

Part	Status
Sangaboard v5	On order
Stepper motors (3x)	On order
Bag of Bits (hardware)	On order
Illumination kit	On order
Tube lens (125–150 mm)	On order
Custom optics STL	In dev.

All parts ordered via Taulab / distributor.
Custom optics module compiled from OpenSCAD source.



Why Custom Optics — The Sensor Utilization Problem

Standard OpenFlexure tube lens (50 mm) was designed for Pi Camera v2 (4.6 mm diagonal).

Our HQ Camera IMX477 has a **9.79 mm diagonal** — **2.1× larger**.

Tube lens	M	EFN	Sensor used
50 mm (std.)	0.261	37.5 mm	~25%
125 mm	0.469	20.9 mm	~85%
150 mm	0.515	19.0 mm	~95%

Calculation:

$$M = \frac{f_t}{f_t + |p|}$$

$$\text{Area ratio} = \left(\frac{4.6}{9.79} \right)^2 \approx 22\%$$

$p = -141.5$ mm (back focal length)

EFN = effective field number

Solution: Replace 50 mm tube lens with **125–150 mm achromatic doublet** (same 12.7 mm diameter). Requires custom optics module STL.



Prospect #1 — Enhancing Data Quality

Equipping experimental base setup with high-quality compounds

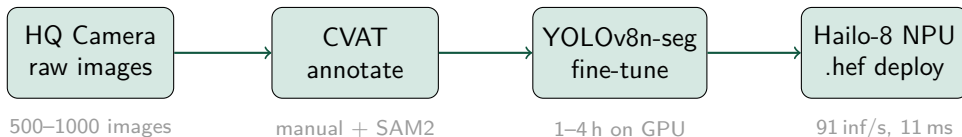
Upgrade	Purpose	Impact
Custom optics module	Match tube lens to IMX477	Full sensor utilization
$\leq 10\times$ RMS objective	Cell-level observation	$\sim 1.3\mu\text{m}$ at $10\times$
Flat-field correction	Even illumination	Consistent intensity
Motorised stage	Sub- μm positioning	Automated time-lapse

Result: Higher quality raw data $\rightarrow$ better annotation $\rightarrow$ better model performance



Prospect #2 — Automated Event Recognition

Event-sensitive cell–cell interaction recognition via NPU inference



Target classes:

- ▶ Osteoblasts
- ▶ Endothelial cells
- ▶ Mineralization zones
- ▶ Membrane boundary

Annotation effort:

Method	Time/image
Manual	5–10 min
SAM2-assisted	1–3 min

500 images $\approx$ 8–42 h
1000 images $\approx$ 17–83 h



Data Storage — Raw Image Estimates

IMX477 sensor: 4056×3040 pixels, 12.3 MP

Format	/image	500	1000
12-bit RAW	18.5 MB	9.2 GB	18.5 GB
16-bit TIFF	24.7 MB	12.3 GB	24.7 GB
JPEG (HQ)	8 MB	4.0 GB	8.0 GB

Time-lapse (48 h):

Interval	Frames	RAW
1 min	2880	53 GB
5 min	576	11 GB
10 min	288	5 GB

Recommendation

Attach **USB 3.0 SSD (256 GB+)** to RPi 5 for raw data collection. Back up to NAS after each session. SD card (32 GB) is *not* sufficient for RAW time-lapse series.



Goal #1 — Biological Insight

Gaining deeper insight into cell–cell interaction dynamics during hydroxyapatite production

- ▶ How does proximity to active osteoblasts affect endothelial cell morphology and migration?
- ▶ Does the rate of calcium deposition change in the presence of endothelial signaling?
- ▶ Can we identify temporal patterns — do interactions follow predictable phases?
- ▶ What morphological markers precede mineralization events?

Measurement approach:

- ▶ Automated time-lapse imaging (hours to days)
- ▶ AI-assisted segmentation of cell populations and mineralization zones
- ▶ Quantitative tracking: cell count, area, migration velocity, mineralization density



Goal #2 — Autonomous Reproducibility

Building a stable, reproducible experimental platform

Requirement	Solution
Hardware reproducibility	Open-source 3D-printed microscope (CERN-OHL)
Software reproducibility	Ansible-automated deployment (<10 min)
Protocol reproducibility	Scripted sessions via OpenFlexure REST API
Analysis reproducibility	Compiled .hef model — deterministic inference

What this means

Any lab with a 3D printer and ~300 EUR can replicate the entire setup.
Software stack is version-controlled. Imaging protocols are code.
AI models are compiled binaries — identical results on any Hailo-8 NPU.



Timeline

Phase	Deliverable	Status
Experiment design	Bioreactor chip, cell lines, protocol	Established
Hardware base	RPi 5, AI HAT, HQ Camera, OFM server	Operational
Optics + stage	Custom optics module, motorised stage	In progress
Early evaluation	First images on conventional microscope	Current
Dataset collection	500–1000 annotated time-lapse images	Pending
Model training	YOLOv8n-seg fine-tuned on cell data	Pending
Deployment	Real-time inference on Hailo-8 NPU	Pending



Open Questions

- ▶ **Flow rate:** Optimal flow rate ($\mu\text{L}/\text{min}$) for Compartment B?
Depends on chip geometry, pump, and shear stress tolerance of hMEC 1.
- ▶ **Objective:** Exact magnification of scraped objective to be verified —
determines tube lens choice (125 mm vs. 150 mm).
- ▶ **Capture interval:** Optimal time-lapse interval for mineralization dynamics
(5 min? 10 min?)
- ▶ **Training data:** Is a single experiment run sufficient for 500–1000 diverse images,
or are multiple runs needed?



Glossary

Term	Definition
hFOB 1.19	Human fetal osteoblast cell line
hMEC 1	Human microvascular endothelial cell line
Hydroxyapatite	$\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$ — primary bone mineral
OpenFlexure	Open-source 3D-printed microscope (Cambridge)
RMS	Royal Microscopical Society — objective thread standard
Tube lens	Achromat between objective and sensor — sets magnification
YOLOv8n-seg	YOLO v8 nano segmentation — lightweight edge model
Hailo HEF	Compiled neural network format for Hailo NPU
NPU	Neural Processing Unit (here: Hailo-8, 27 TOPS)
CVAT	Computer Vision Annotation Tool (open source)
SAM2	Segment Anything Model 2 — automatic pre-segmentation
C-mount	Camera mount standard, 17.526 mm FFD

Thank you

A rectangular box with a thin black border containing the text "University Logo" centered within it.

University
Logo

University of Salzburg

openflexure.org · YOLOv8 · CVAT